

3.5 Linear motion under a variable force

Further Mechanics · 2026 examination · Concept guide and original exam-style practice

Understand the concept

For variable force, Newton's law is $m \, dv/dt = F(t, x, v)$. If acceleration is a function of position, use $a = v \, dv/dx$, then integrate. The work-energy method $\int F \, dx = \Delta(\frac{1}{2}mv^2)$ can simplify finding speed without time.

Worked example: follow the reasoning

For $F(x) = 2x \, \text{N}$ on $1 \, \text{kg}$ object from $x = 0$ to $2 \, \text{m}$, work $\int_0^2 2x \, dx = 4 \, \text{J}$, so speed from rest is $\sqrt{8} \, \text{m s}^{-1}$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A $2 \, \text{kg}$ particle at rest is pushed by $F = 6x \, \text{N}$ from $x = 0$ to $x = 2 \, \text{m}$. Find speed at $x = 2 \, \text{m}$. [3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Work $\int 6x \, dx = 12 \, \text{J}$ [2]; $\frac{1}{2}(2)v^2 = 12$, $v = \sqrt{12} \approx 3.46 \, \text{m s}^{-1}$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

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<https://www.cambridgeinternational.org/Images/697357-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.